Ulyana Piterbarg

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Education

Massachusetts Institute of Technology 09/2017 - 06/2021

Candidate for B.S. in Mathematics with Computer Science (GPA: 4.9/5.0).
Candidate for Minor in Statistics and Data Science.

Relevant Coursework: *Algorithms for Inference, Statistical Learning and Data Mining, Artificial Intelligence, Measure-Theoretic Probability Theory, Computational Cognitive Science, Computability and Complexity Theory.*

Research

Undergraduate Researcher, Climate Modeling Alliance 05/2020 - 05/2021

- Advised by Prof. Raffaele Ferrari (MIT).
- Experimented with Bayesian and deep learning-based model architectures for uncovering novel parameterizations of large-scale turbulence, evaluating results via uncertainty quantification.

Research Assistant, Computational Cognitive Science Group 09/2018 - 05/2021

- Advised by Prof. Joshua B. Tenenbaum (MIT).
- Conducted experiments studying the cognitive bases of strategy learning and physics-engine representations in humans, analyzing results with Bayesian hierarchical models.
- Developed novel model-free, hierarchical reinforcement learning (RL) architectures for fluid manipulation tasks, able to adaptively simulate particle-based physical dynamics via graph networks.

Summer Research Fellow, Laboratory of Computational Neuroscience 06/2018 - 09/2018

- Advised by Prof. Wulfram Gerstner (École polytechnique fédérale de Lausanne).
- Investigated the versatility of a hybrid deep reinforcement learning (RL) planning algorithm in dynamic and partially-observable environments, evaluating response to large-scale environmental changes.
- Employed results of analysis to boost the RL planning algorithm's performance in stochastic environments.

Technical Assistant, MIT Lincoln Laboratory Space Systems and Controls Division 01/2018 - 05/2018

- Advised by Dr. Lulu Liu (Harvard University).
- Performed proof-of-concept system analytics for a novel adaptive optics system using idealized turbulence models as well as true turbulence profiles from astronomical sites all over the world.

Research Assistant, Clark Laboratory 06/2016 - 09/2016

- Advised by Prof. Damon Clark (Yale University).
- Engineered deep learning models for feature extraction and algebraic characterization of modulatory properties of central pattern generator (CPG) biological neural circuits in *D. melanogaster* (fruit flies).

Industry

Investment Associate Intern, Bridgewater Associates 07/2020 - 08/2020

- Studied the macroeconomic effects of the SARS-CoV-2 pandemic on financial markets around the world.

Software Engineering Intern, Spell AI 01/2019 - 02/2019

- Developed several features for a deep learning/ML development platform via full-stack engineering.
- Provided expertise with Bayesian optimization algorithms for hyperparameter tuning.

Design Intern, American Museum of Natural History (AMNH) 05/2017 - 08/2017

- Researched and aided in the development of the AMNH exhibition "Our Senses," visited by millions of people from around the world during its showing from Nov 2017 to Jan 2019.

Publications

1. KR. Allen, KA. Smith, **U. Piterbarg**, R. Chen, JB. Tenenbaum: "Abstract strategy learning underlies flexible transfer in physical problem solving." *42nd Annual Virtual Meeting of the Cognitive Science Society (CogSci 2020).*

Poster Sessions

1. **U. Piterbarg**. *Optimizing Parameterizations of Turbulent Planetary Flows for Climate Modeling with Machine Learning.* MIT SuperUROP Showcase (2020).
2. KR. Allen, KA. Smith, **U. Piterbarg**, R. Chen, JB. Tenenbaum: *Abstract strategy learning underlies flexible transfer in physical problem solving.* "Developing a Mind: Learning in Humans, Animals, and Machines," 42nd Annual Virtual Meeting of the Cognitive Science Society (CogSci 2020).
3. **U. Piterbarg**, J. Brea. *Investigating the Efficacy of Option-Conditional Value Prediction in Reinforcement Learning.* Life Sciences SRP, École polytechnique fédérale de Lausanne (2018).

Projects

1. *A Bayesian Approach to Modeling Infection-Based Social Distancing in the SARS-CoV-2 Pandemic*: Employed Hamiltonian Monte Carlo (HMC) to derive estimates for fundamental epidemiological parameters associated with SARS-CoV-2 disease transmission and to compare the successes of different countries' infection-based social distancing policies throughout the early stages of the pandemic.
2. *Exploring strategy learning in the 'Tools Environment'*: Ran a pilot set of experiments evaluating the `Tools Environment` as a candidate for computational study of the cognitive processes supporting strategy learning and as a benchmark environment for few-shot RL, using a hierarchical Bayesian model to analyze and classify the physical prediction strategies learned by subjects.

Professional Service

Teaching and Grading

Grading Assistant, Fundamentals of Statistics (18.650), MIT	2020
Teaching Assistant, Computational Cognitive Science (6.804/9.66), MIT	2019

Outreach

Representative, Council for Math Majors (CoMM)	2020-2021
Mentor, Undergraduate Society of Women in Math (USWIM)	2020-2021
Member, MIT Society of Women Engineers (SWE)	2017-2021

Skills

Programming: Python, Julia, MATLAB, JavaScript, JAVA, goLang, LaTeX.
Packages and Libraries: scikit-learn, pytorch/tensorflow, pymc3, MuJoCo.
Languages: English (native/fluent), French (Delf B2 Certification), Russian (native/fluent).

Honors and Awards

MIT Mathematics Directed Reading Program (DRP) Scholar	2020
MIT Quest for Intelligence Undergraduate Research and Innovation Scholar	2020
MIT January Scholar in France	2020
EPFL Summer Research Program Fellow	2018
AMNH Design for Science Communication Student	2017
National Merit Scholar	2017
Moody's Math Modeling (M^3) Challenge Finalist	2016